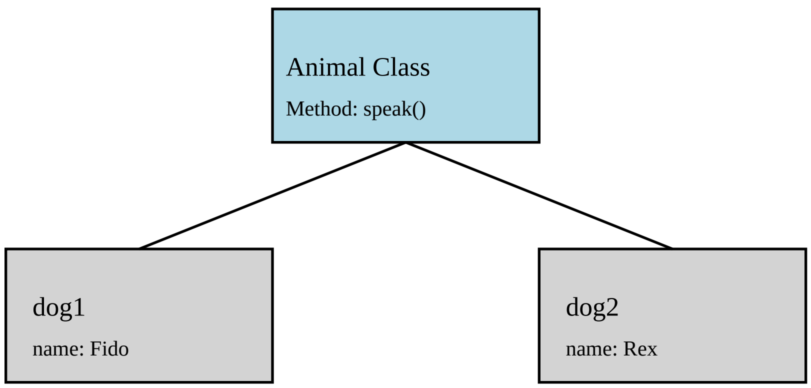
Understanding ECMAScript Inheritance (vs Class-Based OOP)

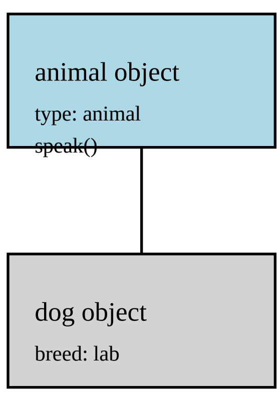
■ Part 1: Class-Based OOP (Traditional Languages like Java/C++)

In class-based languages, each instance carries its own state (like `name`), while methods like `speak()` are defined once in the class and shared by all instances. Inheritance passes down structure and behavior but not actual instance data.



■■ Part 2: ECMAScript Model (Prototype-Based)

JavaScript uses objects as the fundamental building blocks. Objects can carry both state and methods, and inheritance works via prototype chains. This means objects can inherit structure, behavior, and even default state.



■ Comparison Table

Concept	Class-Based Languages	ECMAScript (JavaScript)
State location	In instances	In objects and inherited
Method location	In class (shared)	In objects and inherited
Inheritance type	Structure & behavior	Structure, behavior, & state
Mechanism	Class hierarchy	Prototype chain